

Arian Haghparast

Machine Learning and AI Engineer

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GitHub LinkedIn Google Scholar

Machine learning engineer and researcher focused on retrieval, model evaluation, and ML reliability. Develops and deploys RAG applications and builds Python tools for debugging, data leakage detection, and reproducible experimentation, with experience in deep learning and reinforcement learning.

Technical Skills

ML: Python, PyTorch, TensorFlow, scikit-learn, NumPy, pandas; reinforcement learning, Transformers, graph neural networks.

AI systems: RAG, hybrid retrieval, MongoDB Atlas, model evaluation, experiment tracking, regression testing, caching.

Engineering: SQL, Git, GitHub Actions, JavaScript/TypeScript, Next.js, React, Node.js, Vercel.

Experience

Research Assistant

Jan 2024 – Present

Data Mining Lab · York University

- Built Python pipelines that turn large-scale raw datasets into training-ready inputs and reproducible model evaluations, covering ingestion, cleaning, and feature engineering.
- Developed deep learning and reinforcement learning models for decision-making, data compression, and similarity search using PyTorch and TensorFlow.
- Ran cloud-based GPU training and controlled experiments to compare model performance and isolate the effects of individual design choices.
- Automated experiment reporting to compare model quality, memory use, and failure cases across runs; evaluated robustness to changing data and operating conditions.
- Developed methods that learn continuously from incoming data without retaining past training samples, allowing adaptation while keeping memory use bounded.

AI Engineer, RAG Systems

Sep 2025 – Apr 2026

Research Society at York (RSAY) · York University student organization

- Built and deployed a retrieval-augmented generation assistant to help students find research opportunities, with source citations and unsupported-answer guardrails.
- Reduced cold retrieval latency by 82% using conditional query rewriting and LRU embedding caching; instrumented retrieval stages to identify bottlenecks.
- Combined vector search and BM25 with Reciprocal Rank Fusion in MongoDB Atlas, achieving 55 ms median search latency in retrieval benchmarks.
- Built a labeled evaluation set to measure retrieval quality; benchmarked cross-encoder reranking and kept it optional based on its quality and latency tradeoff.

Education

York University · Lassonde School of Engineering

2026

BSc, Computer Science · GPA: 3.96/4.00

Lassonde Undergraduate Research Award; Annual Continuing Student Scholarship; Dean's Honour Roll.

Selected Projects

MLForensics · ML debugging and regression testing (alpha)

- Built a Python toolkit to investigate ML regressions by comparing metrics and errors alongside changes in code, data, and resource use.
- Implemented run replay, failure minimization, and tensor tracing to isolate model behavior changes and create smaller reproducible cases.
- Added statistical comparisons and CI checks to flag regressions across repeated runs.

Leakproof · ML data leakage detection

- Built a Python tool that flags data leakage and preprocessing mistakes in scikit-learn and pandas workflows.
- Combined static code analysis, dataset audits, and optional runtime probes for train/test overlap and group, temporal, and target leakage.
- Integrated checks into GitHub Actions and generated terminal, JSON, Markdown, and SARIF reports.

Vizir · Interactive data exploration

- Built notebook views and shareable offline HTML reports for data quality and model error analysis.
- Added classification and regression diagnostics with adjustable thresholds and linked selections across data subsets.

Selected Publications

PathletRL++: Optimizing Trajectory Pathlet Extraction and Dictionary Formation via Reinforcement Learning · Co-first author, ACM TSAS, 2026.

Network-Constrained Policy Optimization for Adaptive Multi-Agent Vehicle Routing · Co-first author, ACM TSAS, 2026.

Robot Wheelchair Convoys for Assistive Human Transportation · Co-author, IEEE Smart Mobility, 2024.

Additional Experience

Research Assistant · Elder Laboratory, York University

Sep – Dec 2023

Built a Node.js/WebSockets backend for real-time dataset access.